CodeAlpha

Civil Engineering & Structural Design Internship

Complete submission - Tasks 1, 2, 3 and 4

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Project	G+1 Residential Building - planning, analysis, 3D model and seismic design
Tools / codes	AutoCAD DXF, IS 456, IS 875, IS 1893, IS 13920
Submitted	October 2026

Contents

Task	Title	Deliverable
1	Residential Building Floor Plan (G+1)	Ground & first floor plans (PDF) + CAD file (DXF)
2	Structural Load Analysis	Report + calculation sheet (slab, beam, column, wind, earthquake)
3	3D Modeling & Visualization	3D model (DXF) + 5 rendered images
4	Mini Project - Earthquake-Resistant Building Design	Report with diagrams and charts

All four tasks are based on one building so that the architectural plan, the structural design, the 3D model and the seismic study are consistent with each other. Values that depend on the site (seismic zone, wind speed, soil) are stated as assumptions and should be confirmed for a real project.

TASK 1 - Residential Building Floor Plan

G+1 residential building | AutoCAD DXF + PDF export

1. Design brief

A two-storey (Ground + 1) single-family house has been planned on a **15 m x 14 m (210 m²)** plot with the entrance facing south (plan north is up). The building is a compact rectangular RCC framed structure with 230 mm brick external walls and 115 mm internal partitions, a 3.0 m floor-to-floor height and a 450 mm plinth. The layout follows a simple, regular grid (5.0 + 5.5 m bays in X, 4.5 + 4.5 m bays in Y), which keeps the structural system economical and earthquake-friendly (see Tasks 2 and 4).

Planning principles used: (i) living and dining zone at the entrance side with an open link between them; (ii) kitchen next to the dining room and served by an external window for cross-ventilation; (iii) one bedroom and a common toilet on the ground floor for elderly family members; (iv) private zone (three bedrooms, study/lounge) on the first floor; (v) a central dog-legged staircase that also acts as a stiff core; (vi) every habitable room has at least one external window, and all wet areas (toilets, kitchen) are stacked one above the other to simplify plumbing.

2. Accommodation and area statement

Floor	Room	Size (clear, m)	Area (m²)	Floor	Room	Size (clear, m)	Area (m²)
GF	Living Room	4.83 x 4.33	20.9	FF	Master Bedroom	4.83 x 4.33	20.9
GF	Dining	5.33 x 4.33	23.1	FF	Bedroom-3	5.33 x 4.33	23.1
GF	Bedroom-1	3.83 x 4.33	16.6	FF	Bedroom-2	3.83 x 4.33	16.6
GF	Passage	1.38 x 2.38	3.3	FF	Lobby	1.38 x 2.38	3.3
GF	Toilet	1.38 x 1.83	2.5	FF	Toilet	1.38 x 1.83	2.5
GF	Staircase	2.08 x 4.33	9.0	FF	Staircase	2.08 x 4.33	9.0
GF	Kitchen	2.63 x 4.33	11.4	FF	Study / Lounge	2.63 x 4.33	11.4
				FF	M. Toilet	1.63 x 1.68	2.7
	Carpet area GF		**86.7**		**Carpet area FF**		**89.5**

Built-up area per floor (outer wall face) = 10.73 x 9.23 = **99.0 m²**; total built-up area (2 floors) = **198.1 m²** plus porch 3.0 m² and balcony 4.8 m². Plot coverage = 47% (ground coverage - verify the permissible limit with local building bye-laws). All habitable rooms exceed the commonly used minimum of 9.5 m² (min. width 2.4 m); kitchen and toilets also meet the usual minimum sizes.

3. Door and window schedule

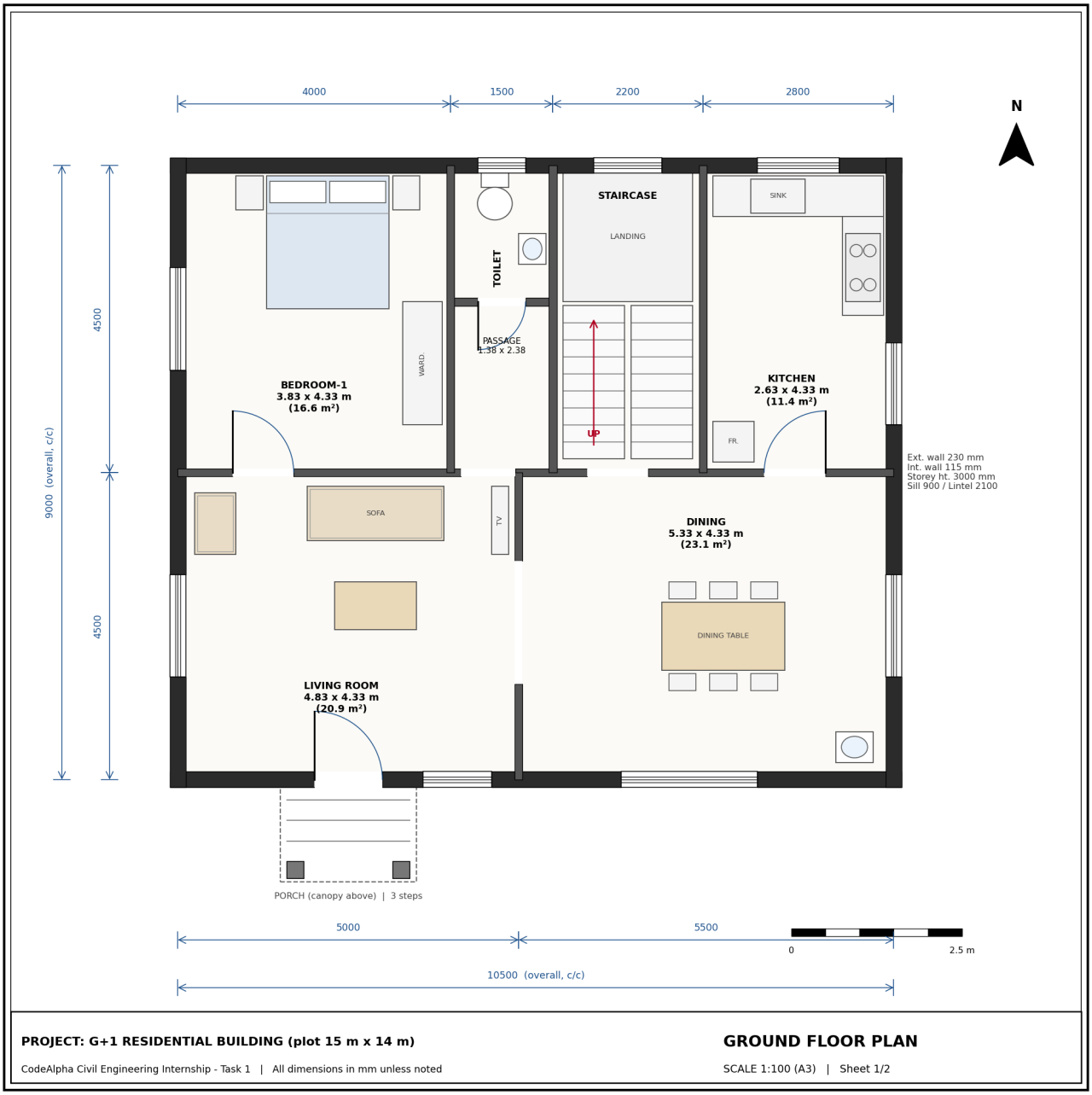
Mark	Width (mm)	Height (mm)	Location	Type / finish
D1	1000	2100	Main / balcony door	Teak wood panelled
D2	900	2100	Bedroom / kitchen door	Flush door, wooden frame
D3	750	2100	Toilet door	PVC / flush door

Mark	Width (mm)	Height (mm)	Sill (mm)	Location	Type / finish
W1	1500	1200	900	Bedroom / living	Aluminium sliding, 5 mm glass
W2	1800	1200	900	Dining / master bedroom	Aluminium sliding, 5 mm glass
W3	1200	1200	900	Kitchen / study	Aluminium sliding, 5 mm glass
V1	700	600	1500	Toilet ventilator	Louvered, frosted glass

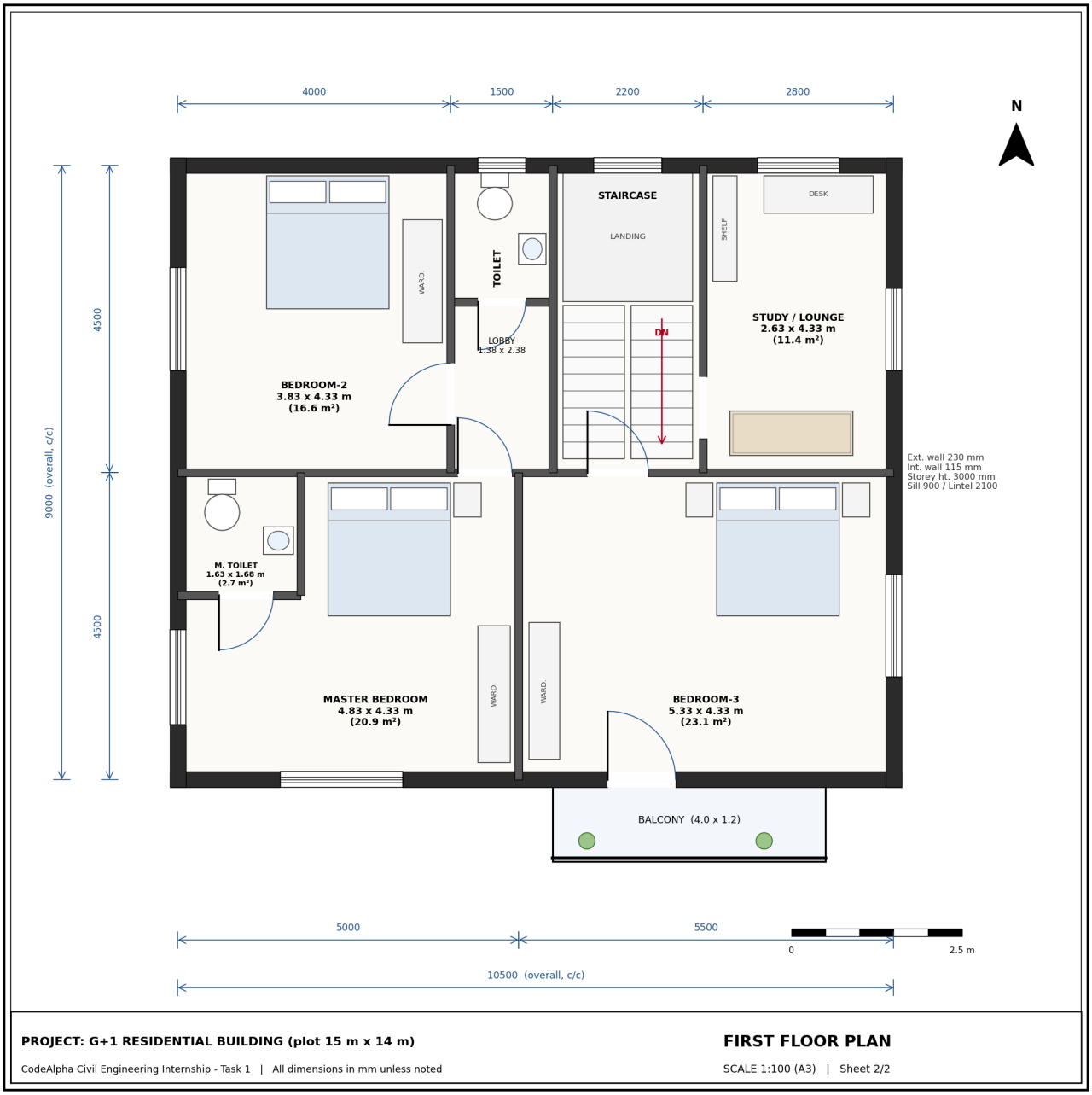
4. Drawing notes and specifications

- Scale 1:100 on A3; all dimensions in millimetres (DXF units = mm, layers: WALL-EXT, WALL-INT, DOOR, WINDOW, FURNITURE, STAIR, TEXT, DIMENSION).
- Walls: 230 mm external and 115 mm internal brick masonry in cement mortar 1:6, plastered both sides. Lintel level 2100 mm, window sill 900 mm (ventilator sill 1500 mm).
- Staircase: dog-legged, 20 risers of 150 mm and treads of 250 mm, flight width 0.9 m (increase to 1.0 m if required by local bye-laws); head-room over the stair > 2.2 m.
- Porch 2.0 x 1.5 m with RCC canopy; first-floor balcony 4.0 x 1.2 m with 1.0 m railing; toilets have ventilators (V1), master toilet an exhaust fan.
- Structure: RCC frame (M25, Fe500), 160 mm slabs, 230 x 500 beams, 300 x 300 columns (Task 2).

Ground Floor Plan - scale 1:100 (A3 sheet reduced to fit A4)



First Floor Plan - scale 1:100 (A3 sheet reduced to fit A4)



TASK 2 - Structural Load Analysis

Slab, beam, column + wind and earthquake loads | IS codes

1. Objective and structural system

This report presents the load calculation and analysis of the main structural elements of the G+1 building planned in Task 1: a two-way floor **slab S1** (living-room panel 4.5 x 5.0 m), an edge **beam B1** (span 5.0 m) and a ground-floor **column C1**, followed by basic-level **wind** and **earthquake** loads on the whole building. Design uses the limit-state method of IS 456:2000 with **M25** concrete (f_{ck} = 25 N/mm²) and **Fe500** steel (f_y = 500 N/mm²). Loads follow IS 875 (Parts 1-3) and IS 1893 (Part 1):2016; detailing follows SP 16 and IS 13920:2016. All assumptions are stated so that the values can be changed in a software model (STAAD.Pro / ETABS).

2. Design loads

Load	Value	Basis
Self weight of RCC	25 kN/m³	IS 875 Part 1
Slab self weight (160 mm)	4.0 kN/m²	0.16 x 25
Floor finish + screed + ceiling plaster (dead)	1.0 kN/m²	Assumed
Brick masonry (230 mm, plastered)	20 kN/m³ = 11.5 kN/m per 2.5 m wall	IS 875 Part 1
Live load - residential rooms	2.0 kN/m²	IS 875 Part 2
Live load - accessible roof	1.5 kN/m²	IS 875 Part 2
Design combinations	1.5(DL+LL); 1.2(DL+LL±EL); 1.5(DL±EL); 0.9DL±1.5EL	IS 456 Table 18

Total service load on floor slab w = 4.0 + 1.0 + 2.0 = **7.0 kN/m²**; factored load w_u = 1.5 x 7.0 = **10.5 kN/m²**.

3. Slab S1 - two-way slab (two adjacent edges discontinuous)

Effective spans l_x = 4.5 m, l_y = 5.0 m, so l_y/l_x = 1.11 < 2 and the slab acts in two-way action. Bending-moment coefficients are taken from IS 456 Table 26 (case 4) and interpolated: α_x(-ve) = 0.0538, α_x(+ve) = 0.0406, α_y(-ve) = 0.047, α_y(+ve) = 0.035. Overall depth D = 160 mm, effective depth d = 135 mm (span/depth = 33.3 < 26 x 1.4, OK).

Section	Mu (kN·m/m)	Ast req. (mm²/m)	Reinforcement	Ast prov. (mm²/m)
Short span, mid (+ve)	8.62	192 (min)	10φ @ 300 c/c	262
Short span, support (-ve)	11.43	201	10φ @ 300 c/c	262
Long span, mid (+ve)	7.44	192 (min)	8φ @ 260 c/c	193
Long span, support (-ve)	9.99	192 (min)	8φ @ 260 c/c	193

Moment capacity check: $M_{u,lim} = 0.138\, f_{ck}\, b\, d^2$ = 62.9 kN·m >> 11.4 kN·m (safe, under-reinforced). Minimum steel 0.12% bD = 192 mm²/m governs. Shear: $V_u = w_u l_x/2$ = 23.6 kN/m, τ_v = 0.17 N/mm² < τ_c = 0.32 N/mm² - no shear reinforcement needed. Torsion steel (0.75 A_{st} over l/5 at the two discontinuous corners) and 2 x 10φ extra top bars at edges are to be added as per IS 456 cl. D-1.8.

4. Beam B1 - 230 x 500 mm, simply-supported span 5.0 m (edge beam)

Loads on beam (factored): slab load transferred as equivalent UDL = $w_u l_x/6 \times [3 - (l_x/l_y)^2]$ = **17.2 kN/m**; self-weight = 0.23 x 0.5 x 25 x 1.5 = **4.31 kN/m**; 230 mm brick wall (2.5 m high) = 0.23 x 2.5 x 20 x 1.5 = **17.3 kN/m**. Total w_u = **38.8 kN/m**. Hence $M_u = w_u L^2/8$ = **121.3 kN·m** and $V_u = w_u L/2$ = **97.0 kN** (conservatively taken as simply supported; continuity will reduce the sagging moment).

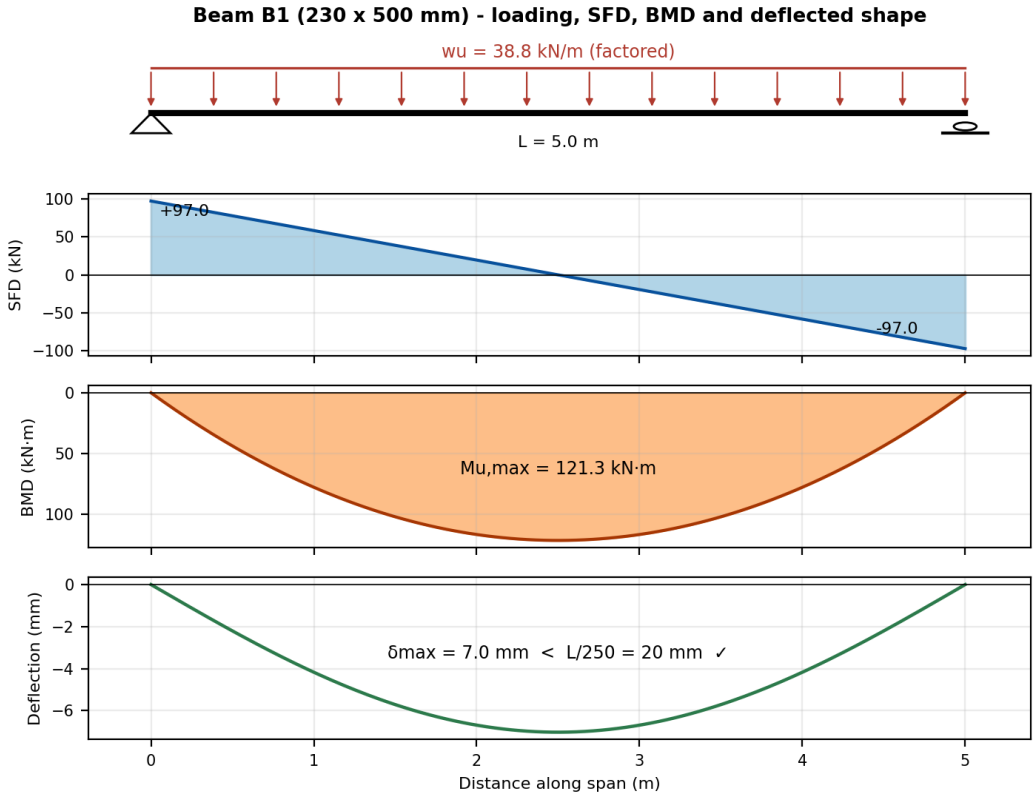


Fig. 1 Loading, shear-force diagram, bending-moment diagram and deflected shape of beam B1

Flexure: $M_{u,lim} = 0.138\, f_{ck}\, b\, d^2 = 167.9$ kN·m > $M_u = 121.3$ kN·m, so a singly reinforced section is adequate. A_{st} required = **698 mm²** → provide **4-16φ** ($A_{st} = 804$ mm², $p_t = 0.76\%$), min. steel $0.85bd/f_y = 180$ mm² OK. **Shear:** $\tau_v = 0.92$ N/mm² > $\tau_c = 0.57$ N/mm² (< $\tau_{c,max} = 3.1$), so stirrups are needed: $V_{us} = 36.4$ kN; spacing from strength = 552 mm, but maximum allowed = 300 mm → provide **2-legged 8φ @ 300 mm c/c** (closer, 8φ @ 100-150 mm, within 2d of supports for ductile detailing). **Serviceability:** span/depth = 10.9 < 20 x 1.3 = 26 OK; elastic deflection with cracked stiffness ($I_{eff} \approx 0.5\, I_g$) = 7.0 mm < L/250 = 20 mm OK.

5. Column C1 - 300 x 300 mm (corner/edge column, G+1)

Tributary area = 4.5 x 5.0 = 22.5 m². Axial service load from roof, first floor, beams, brick walls, parapet and self weight is **P = 645 kN**; factored $P_u = 1.5P$ = **968 kN** (detailed in the calculation sheet). Unsupported length 2.5 m, effective length 0.65 x 3.0 = 1.95 m, $l_{ex}/D = 6.5 < 12$ → short column; minimum eccentricity e_{min} = 15 mm ≤ 0.05D = 15 mm, so IS 456 cl. 39.3 applies: $P_u = 0.4\, f_{ck}\, A_c + 0.67\, f_y\, A_{sc}$. With **4-16φ** ($A_{sc} = 804$ mm² = 0.89% > 0.8%), capacity = **1161 kN** > 968 kN (utilisation 83%). Lateral ties 8φ @ 250 mm c/c (least of 300, 16 x 16, 300 mm).

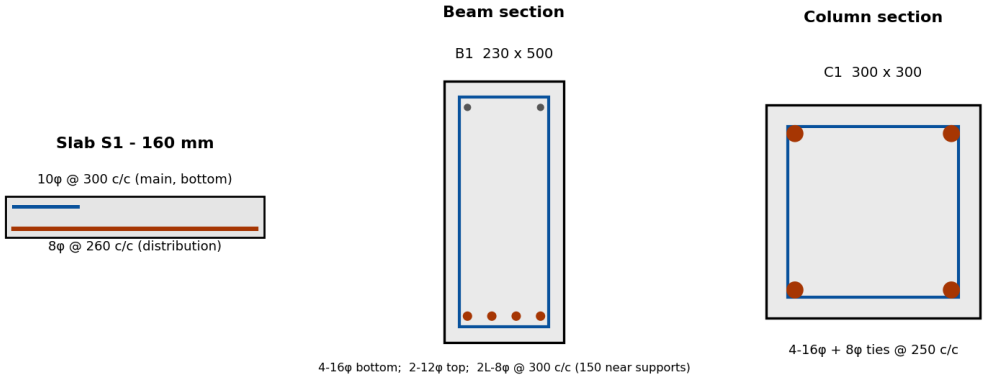


Fig. 2 Reinforcement details: slab strip, beam B1 and column C1

6. Lateral loads - wind and earthquake (basic level)

Wind (IS 875 Part 3): basic wind speed $V_b = 39$ m/s (assumed), $k_1 = k_2 = k_3 = k_4 = 1.0$ → $V_z = 39$ m/s; design pressure $p_z = 0.6V_z^2 = 0.913$ kN/m². With net pressure coefficient $C_p = 1.0$ (windward +0.7, leeward -0.3) on the wind-facing wall 10.73 m x 7.45 m = 79.9 m²: **F_w = 73.0 kN**, overturning moment = 272 kN·m.

Earthquake (IS 1893 Part 1:2016): Zone IV (Z = 0.24), I = 1.0, R = 5 (special moment frame), medium soil. Approximate period $T_a = 0.075h^{0.75} = 0.29$ s → $S_a/g = 2.5$. $A_h = (Z/2)(I/R)(S_a/g) =$ **0.060**. Seismic weight (DL + 25% LL): $W_1 = 1478$ kN, $W_2 = 1277$ kN, W = 2755 kN. Base shear $V_B = A_h W =$ **165.3 kN**, distributed as $Q_i = V_B W_i h_i^2 / \Sigma W_j h_j^2$: **$Q_1$ = 37.1 kN** (first floor), **Q_2 = 128.2 kN** (roof).

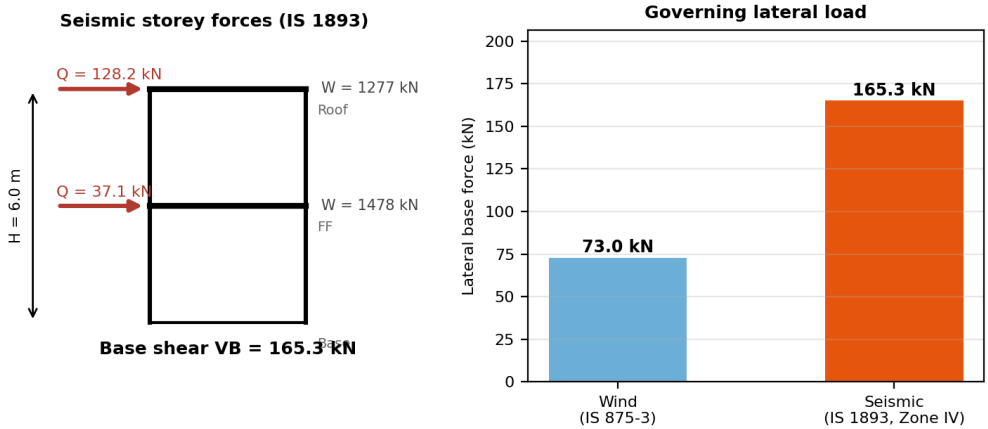


Fig. 3 Storey seismic forces and comparison of wind and seismic base force

7. Structural behaviour

Gravity load path: the slab carries floor loads in bending in both directions and passes them to the supporting beams as triangular (short side) and trapezoidal (long side) loads; beams carry these in bending and shear to the columns, which work mainly in axial compression and deliver the load to the footings and soil. The beam has a maximum sagging moment at mid-span (121 kN·m) and maximum shear at the supports (97 kN), which is why tension steel is concentrated at the bottom of the mid-span and stirrups are closer near the supports.

Lateral behaviour: the seismic base shear (165 kN) is about 2.3 times the wind force (73 kN), so **earthquake governs** the lateral design of this building. The rigid floor slabs act as diaphragms distributing the inertia forces to the frames, and the frames resist them through bending of beams and columns (sway). Approximately 78% of the base shear acts at the roof level because of the higher mass-height product, so the upper storey columns and the beam-column joints need ductile detailing (IS 13920). Storey drift must be limited to 0.004h = 12 mm (IS 1893 cl. 7.11.1.1) - to be verified in the STAAD/ETABS frame model.

8. Conclusion

The adopted sizes - slab 160 mm, beam 230 x 500 mm with 4-16φ bottom bars, column 300 x 300 mm with 4-16φ - satisfy the strength and serviceability checks under gravity loads, and the structure is governed by earthquake rather than wind for lateral loading. The next step for a real project would be a 3D frame analysis with seismic load combinations and verification of the strong-column/weak-beam requirement (see Task 4).

CALCULATION SHEET - Task 2

Step-by-step design calculations
(limit-state method, IS 456:2000)

A. Slab S1 (4.5 m x 5.0 m, D = 160 mm, d = 135 mm, cover 20 mm)

Step	Calculation	Result	Ref.
1	Dead load: slab 0.16 x 25 + finish 1.0	5.0 kN/m²	IS 875-1
2	Live load (residential)	2.0 kN/m²	IS 875-2
3	w = DL + LL = 5.0 + 2.0	7.0 kN/m²	
4	w_u = 1.5 x 7.0	10.5 kN/m²	IS 456 T.18
5	l_y/l_x = 5.0/4.5	1.111 (two-way)	cl. 24.4
6	$M_x(-)$ = 0.0538 x 10.5 x 4.5²	11.43 kN·m/m	Table 26
7	$M_x(+)$ = 0.0406 x 10.5 x 4.5²	8.62 kN·m/m	Table 26
8	$M_y(-)$ = 0.047 x 10.5 x 4.5²	9.99 kN·m/m	Table 26
9	$M_y(+)$ = 0.035 x 10.5 x 4.5²	7.44 kN·m/m	Table 26
10	$M_{u,lim}$ = 0.138 x 25 x 1000 x 135² / 10^6	62.9 kN·m (OK)	Annex G
11	A_{st}: 0.87 f_y A_{st} d [1 - $A_{st}f_y/(b\,d\,f_{ck})$] = M_u	see slab table, Task 2 §3	Annex G
12	$A_{st,min}$ = 0.12% x 1000 x 160	192 mm²/m	cl. 26.5.2.1
13	Spacing s = 1000 x a_φ / A_{st}; max = min(3d, 300)	10φ @ 300; 8φ @ 260	cl. 26.3.3
14	V_u = 10.5 x 4.5 / 2; $\tau_v = V_u/(b\,d)$	23.6 kN; 0.17 N/mm²	cl. 40.1
15	τ_c (p_t = 0.19%, M25)	0.32 N/mm² > τ_v (safe)	Table 19

B. Beam B1 (b = 230 mm, D = 500 mm, span L = 5.0 m)

Step	Calculation	Result	Ref.
1	Slab load (trapezoidal → UDL): $w_u l_x/6 \times (3 - (l_x/l_y)^2)$ = 10.5 x 4.5/6 x (3 - 0.81)	17.25 kN/m	IS 456 / std.
2	Self weight: 1.5 x 0.23 x 0.50 x 25	4.31 kN/m	
3	Wall: 1.5 x 0.23 x 2.5 x 20	17.25 kN/m	IS 875-1
4	w_u = 1 + 2 + 3	38.81 kN/m	
5	$M_u = w_u L^2/8$ = 38.81 x 5² / 8	121.3 kN·m	
6	$V_u = w_u L/2$	97.0 kN	
7	d = 500 - 25 - 8 - 8 (cover, stirrup, half bar)	≈ 460 mm	
8	$M_{u,lim}$ = 0.138 x 25 x 230 x 460² / 10^6	167.9 kN·m > M_u	Annex G
9	A_{st} from quadratic of M_u = 0.87 f_y A_{st} d (1 - $A_{st}f_y/bdf_{ck}$)	698 mm²	Annex G
10	Provide 4-16φ: 4 x 201.1	804 mm² (p_t = 0.76%)	
11	$A_{st,min} = 0.85\,b\,d / f_y$	180 mm² (OK)	cl. 26.5.1.1
12	$\tau_v = V_u/(b\,d)$	0.92 N/mm²	cl. 40.1
13	τ_c (p_t = 0.76%)	0.57 N/mm² ($\tau_{c,max}$ = 3.1)	Table 19/20
14	$V_{us} = V_u - \tau_c bd$	36.4 kN	cl. 40.4
15	$s_v = 0.87\,f_y\,A_{sv}\,d / V_{us}$ (A_{sv} = 100.5 mm²)	552 mm	cl. 40.4
16	$s_{v,max}$ = min(0.75d, 300, $0.87 f_y A_{sv}/0.4b$)	300 mm → provide 300	cl. 26.5.1.5/6
17	Deflection: $\delta = 5\,w_s\,L^4 / (384\,E_c\,0.5\,I_g)$; $E_c = 5000\sqrt{f_{ck}}$ = 25000 MPa, $w_s = w_u/1.5$ = 25.9 kN/m	7.0 mm < 20 mm	cl. 23.2

C. Column C1 (300 x 300 mm, M25 / Fe500)

No.	Axial load component	P (kN)
1	Roof slab DL (5.5 kN/m² x 22.5 m²)	123.8
2	Roof slab LL (1.5 kN/m² x 22.5 m²)	33.8
3	First-floor slab DL (5.0 kN/m² x 22.5 m²)	112.5
4	First-floor slab LL (2.0 kN/m² x 22.5 m²)	45.0
5	Beams, 2 floors (2.875 kN/m x 9.5 m x 2)	54.6
6	Brick wall, FF level + GF level (11.5 kN/m x 9.5 m x 2)	218.5
7	Parapet at roof (4.6 kN/m x 9.5 m)	43.7
8	Column self weight (2 storeys)	13.5
	Total service load P	**645.3**
	Factored P_u = 1.5 P	**968.0**

Step	Calculation	Result	Ref.
1	Slenderness l_{ex}/D = 0.65 x 3000 / 300	6.5 < 12 → short	cl. 25.1.2
2	e_{min} = l/500 + D/30 = 2500/500 + 300/30	15 mm ≤ 0.05D = 15 mm	cl. 25.4
3	$A_{sc,min}$ = 0.8% x 90000	720 mm²	cl. 26.5.3.1
4	Provide 4-16φ	804 mm² (0.89%)	
5	$P_{u,cap} = 0.4 f_{ck}(A_g - A_{sc}) + 0.67 f_y A_{sc}$	1161 kN > 968 kN	cl. 39.3
6	Lateral ties: min(300, 16 x 16 = 256, 300) → 8φ @ 250	OK	cl. 26.5.3.2

D. Wind load

Step	Calculation	Result	Ref.
1	V_b (basic wind speed, assumed)	39 m/s	IS 875-3 Fig. 1
2	$V_z = V_b k_1 k_2 k_3 k_4$ (all = 1.0)	39 m/s	cl. 6.3
3	$p_z = 0.6 V_z^2$	913 N/m² = 0.913 kN/m²	cl. 7.2
4	Exposed area A = 10.73 x 7.45	79.9 m²	
5	$F_w = C_p p_z A$ (C_p = 1.0)	73.0 kN	cl. 7.3
6	$M_{ot} = F_w$ x H/2	272 kN·m	

E. Earthquake load (equivalent static method)

Step	Calculation	Result	Ref.
1	Z = 0.24 (Zone IV), I = 1.0, R = 5.0 (SMRF), medium soil		IS 1893 T.3,8,9
2	$T_a = 0.075 h^{0.75} = 0.075 \times 6.0^{0.75}$	0.288 s	cl. 7.6.2
3	S_a/g (0.10 s ≤ T ≤ 0.55 s, medium soil)	2.50	Fig. 2
4	$A_h = (Z/2)(I/R)(S_a/g)$ = 0.12 x 0.2 x 2.5	0.060	cl. 6.4.2
5	W_1: slab 5.0x94.5 + 25%LL 0.25x2.0x94.5 + beams 198 + cols 81 + walls 678	1478 kN	cl. 7.4
6	W_2: slab 5.5x94.5 + beams + ½cols + ½walls + parapet 179	1277 kN	cl. 7.4
7	$W = W_1 + W_2$	2755 kN	
8	$V_B = A_h W$	165.3 kN	cl. 7.6.1
9	$Q_1 = V_B W_1 h_1^2 / \Sigma(W_j h_j^2)$ (h = 3 m)	37.1 kN	cl. 7.6.3
10	$Q_2 = V_B W_2 h_2^2 / \Sigma(W_j h_j^2)$ (h = 6 m)	128.2 kN	cl. 7.6.3
11	Overturning moment = Q_1x3 + Q_2x6	881 kN·m	

Note: the seismic zone (Z = 0.24) and basic wind speed (39 m/s) are assumed for illustration; for a real site the local code/zone map must be used. Values are basic-level hand calculations intended to be cross-checked with STAAD.Pro / ETABS.

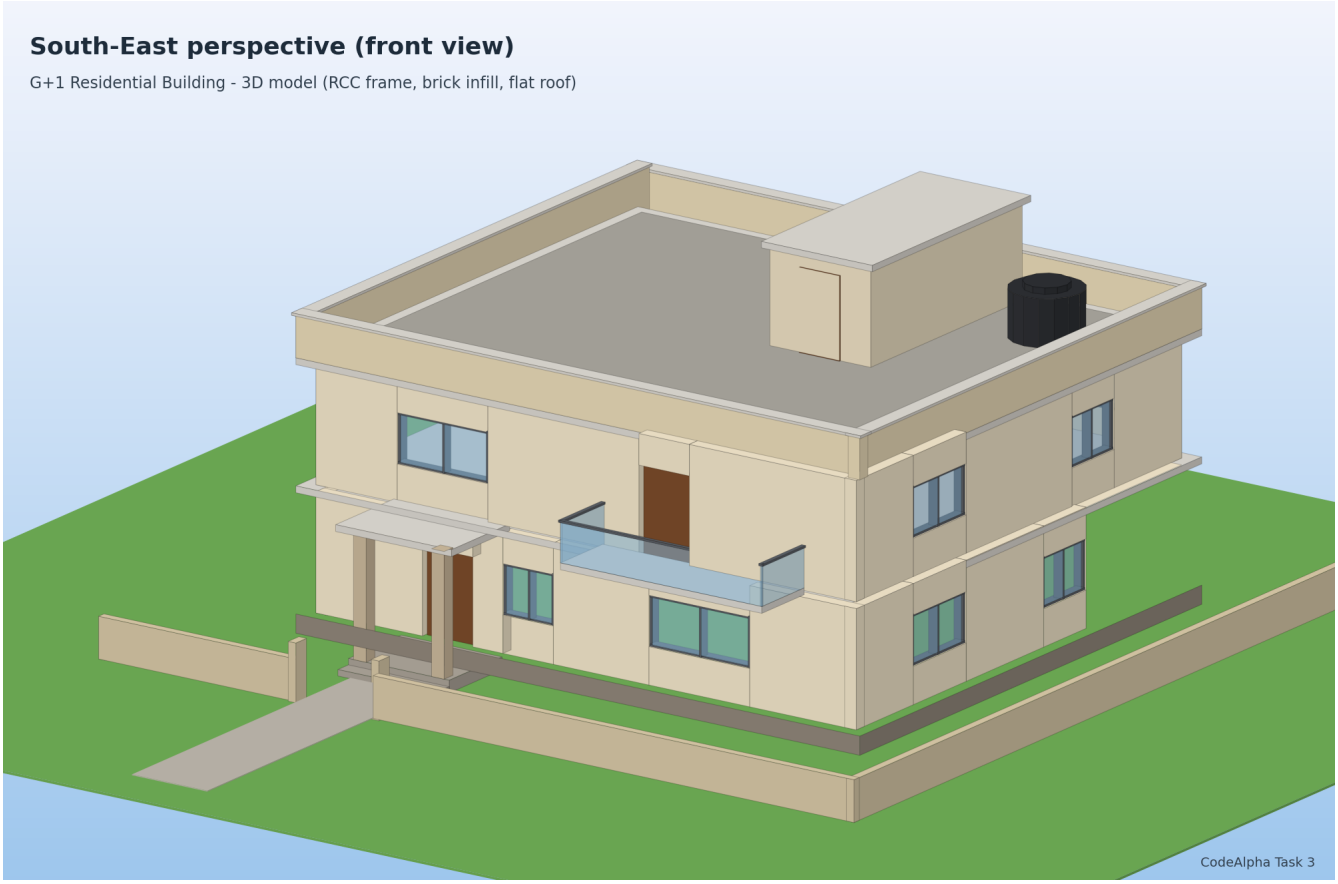
TASK 3 - 3D Modeling & Visualization

G+1 residential house | 3D DXF model + rendered images

1. Model description

A full 3D model of the G+1 house from Task 1 was built to the same dimensions as the 2D plan: RCC plinth, brick walls with real openings for every door and window, floor and roof slabs, parapet, staircase head-room (mumty), overhead water tank, entrance porch with canopy and columns, first-floor balcony with glass railing, compound wall with gate and a garden/path. The model is exported as a **3D DXF** (3DFACE entities on material layers) that can be opened directly in **AutoCAD** and imported into **SketchUp** or **Revit**; materials are applied by layer colour and transparency. The rendered images below are generated from the same geometry.

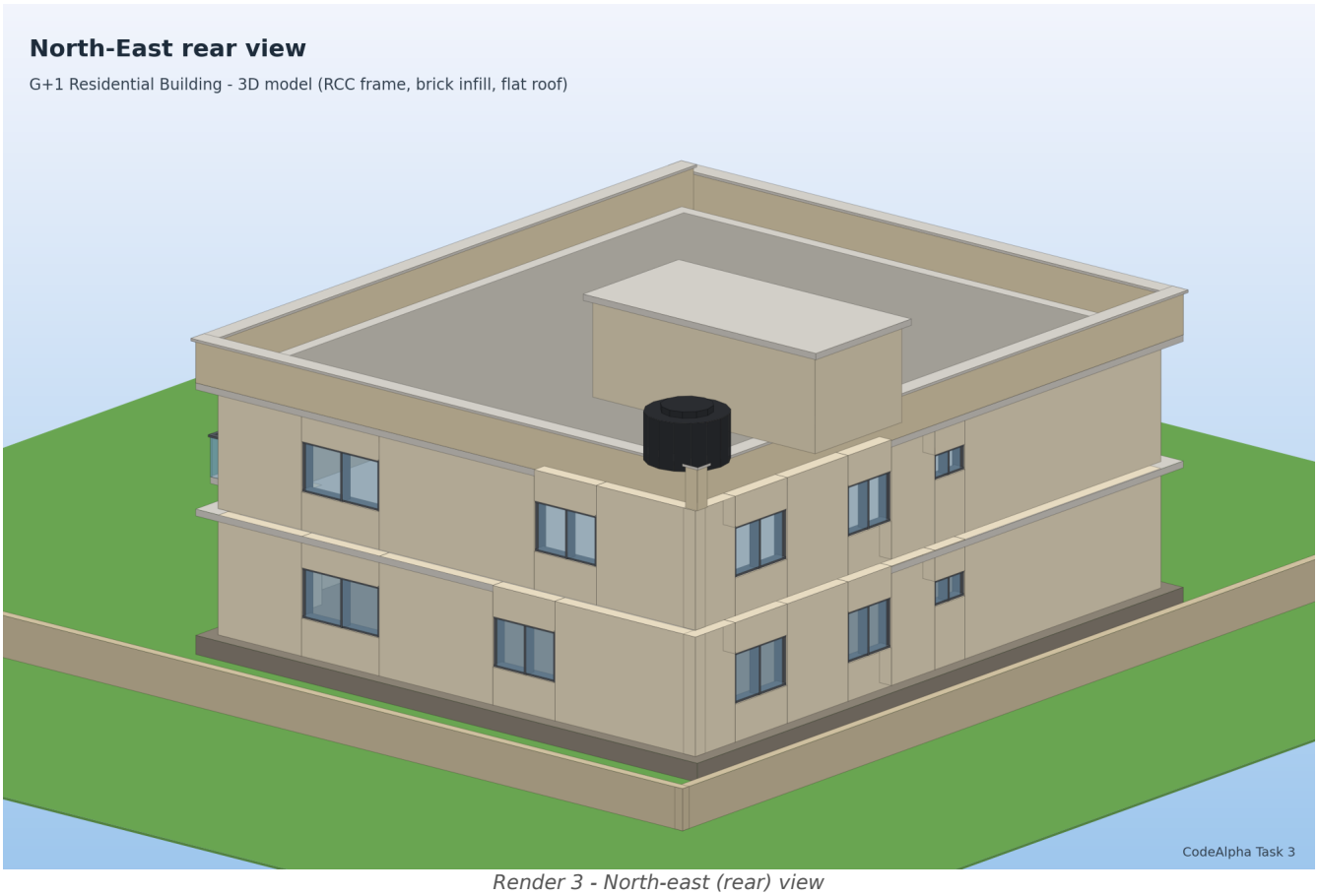
Component	Material / finish	Key dimension
Plinth	Stone-grey cement finish	Height 450 mm
External walls	Cream exterior plaster / paint	230 mm, floor height 3.0 m
Floor and roof slabs	Light-grey concrete, roof finish grey tiles	160 mm, overall height 6.45 m + 1.0 m parapet
Windows	Aluminium frames + blue-tinted 5 mm glass (55% transparent)	Sill 0.9 m, head 2.1 m
Doors	Teak-brown timber	2.1 m high
Balcony railing	Glass panel + steel handrail	1.0 m high, 4.0 x 1.2 m
Porch	RCC canopy on two pillars, three steps	2.0 x 1.5 m
Roof items	Dark-grey PVC water tank, stair mumty	Tank Ø 1.3 m, mumty 2.3 m high
Site	Lawn, paver path, 1.0 m boundary wall	Plot 15 m x 14 m



Render 1 - South-east perspective (main entrance, porch and balcony)



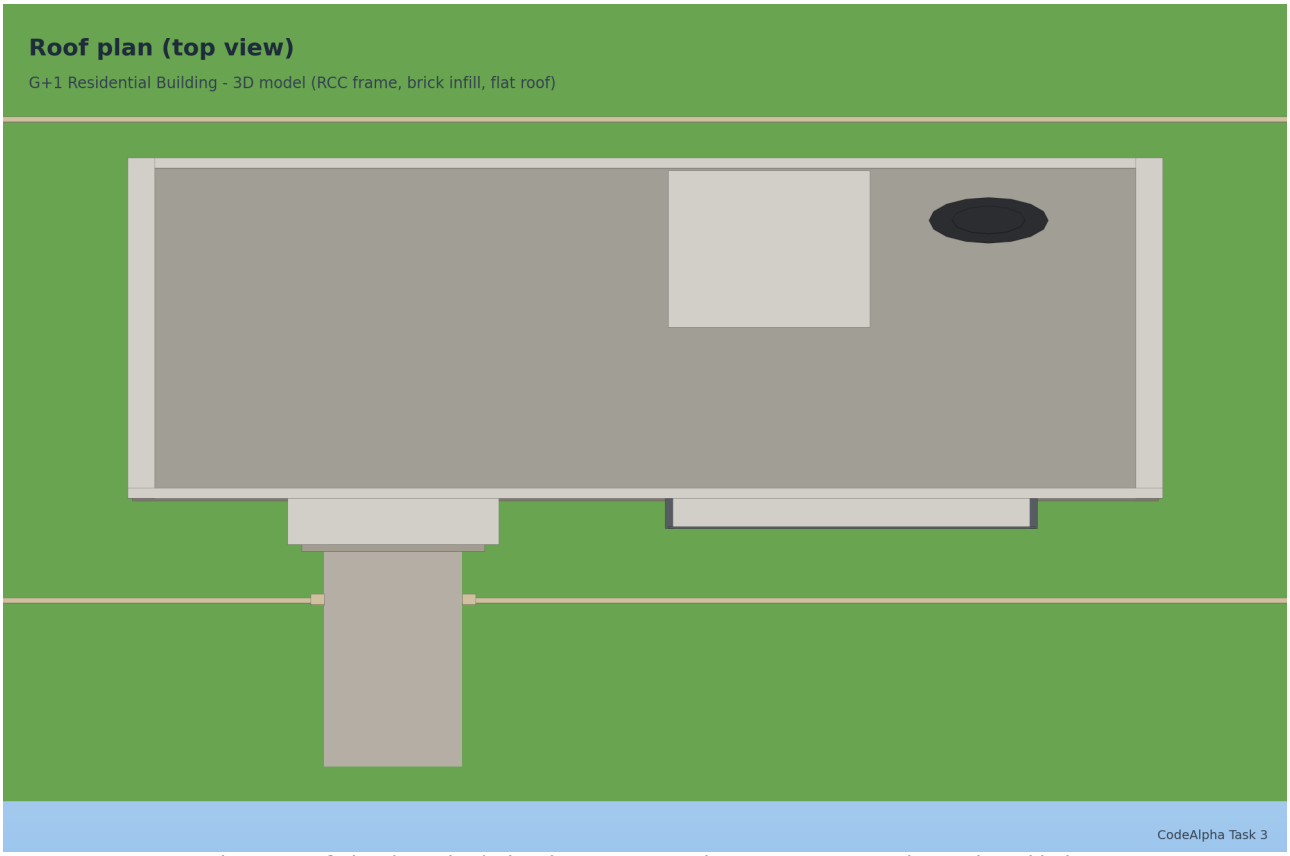
Render 2 - South-west perspective



Render 3 - North-east (rear) view



Render 4 - Front elevation (south)



Render 5 - Roof plan (top view) showing terrace, stair mumty, water tank, porch and balcony

Files submitted: CodeAlpha_Task3_GPlus1_3D_Model.dxf (3D design file) and the five rendered images (PNG) above. To reproduce a photo-realistic render in SketchUp/Revit, import the DXF, assign the named material layers (3D-WALL, 3D-GLASS, 3D-DOOR, ...) and render.

TASK 4 - Mini Project: Earthquake-Resistant Building Design Concepts

Structural design proposal | Case study: G+1 RCC house

Abstract

Earthquakes do not kill people - collapsing buildings do. This mini project explains the concepts that make a building earthquake-resistant: how a structure responds to ground shaking, the design philosophy of modern codes, the importance of a regular configuration, the choice of a lateral load-resisting system and ductile detailing of reinforced concrete. The concepts are then applied to the G+1 RCC house designed in Tasks 1 and 2, for which the seismic base shear, storey forces and detailing requirements are worked out. The report ends with practical recommendations for safe construction in seismically active regions such as Khyber Pakhtunkhwa.

1. Introduction

Earthquakes are caused by the sudden release of strain energy along faults in the earth's crust. Northern Pakistan lies on the collision zone of the Indian and Eurasian plates and has a long record of damaging earthquakes: the 2005 Kashmir earthquake (Mw 7.6) killed roughly 87,000 people in Pakistan, and the 2015 Hindu Kush earthquake (Mw 7.5) caused deaths and widespread house damage across Khyber Pakhtunkhwa. In both events the main cause of casualties was the collapse of non-engineered masonry and poorly detailed concrete buildings.

A building is normally designed for gravity, which is constant and vertical. An earthquake instead imposes **repeated, reversing horizontal inertia forces** that may be as large as a significant fraction of the building's weight. Structures that are economical for gravity alone can be brittle and unsafe under this action. The aim of earthquake-resistant design is therefore not to make a building "earthquake-proof", but to ensure that it **does not collapse** in a rare severe earthquake and **suffers little or no damage** in frequent minor ones. This report is based on IS 1893 (Part 1):2016 and IS 13920:2016; the Building Code of Pakistan - Seismic Provisions (BCP-SP 2007) follows the same principles, and the locally governing zone factor must always be confirmed for a real site.

2. How a building responds to an earthquake

When the ground moves, the base of the building moves with it, but the floors above tend to stay in place because of inertia. The resulting deformation generates an inertia force at each floor equal to **F = m x a** (floor mass times acceleration). Because mass lies mainly at floor levels, the floor slabs, acting as rigid diaphragms, collect these forces and pass them to beams, columns and finally the foundation - the **lateral load path** (Fig. 2). The size of the force depends on the natural period T of the building: short, stiff buildings (T = 0.2-0.5 s) attract the highest accelerations, while tall flexible buildings attract less force but undergo larger displacements (Fig. 1). Four attributes govern the performance of a building: a **simple and regular configuration**, adequate **lateral strength**, adequate **stiffness** (to limit drift and damage) and, above all, **ductility** - the ability to deform well beyond yield without losing strength.

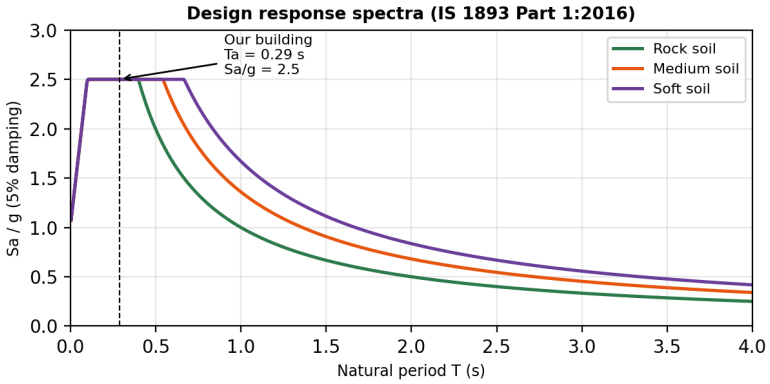
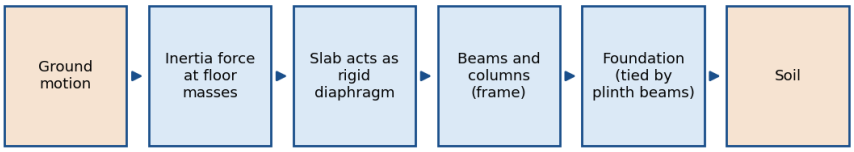


Fig. 1 Design acceleration spectrum: the building in this project (T_a = 0.29 s) sits on the plateau, i.e. it attracts the maximum design acceleration

Lateral load path in an earthquake-resistant building



Continuous load path: every link must be strong enough AND ductile — one weak link governs the whole building

Fig. 2 The seismic load path - a chain that is only as strong as its weakest link

3. Seismic design philosophy

Codes do not design buildings to remain elastic in the strongest shaking, because that would be uneconomical. Instead they use a three-level performance philosophy:

- **Minor (frequent) shaking:** the building should remain essentially undamaged (serviceable).
- **Moderate shaking:** structural damage should be limited and repairable.
- **Severe (rare, maximum considered) shaking:** major damage may occur but the building must **not collapse** - life safety is the priority.

To achieve this economically, the elastic design force is divided by a **response reduction factor R** that credits the ductility and over-strength of the structure: $V_B = A_h W$ with $A_h = (Z/2)(I/R)(S_a/g)$. The benefit of R is available *only if the structure is detailed to be ductile*, which is why detailing is as important as the calculated force. Modern codes also enforce **capacity design**: members are proportioned so that the desired ductile mechanism forms first - beams yield in flexure before columns (strong-column/weak-beam), and flexural yielding occurs before brittle shear or bond failure. Storey drift under design forces is limited to 0.4% of storey height.

4. Configuration - the cheapest form of seismic safety

Damage surveys repeatedly show that the geometry of a building matters more than small differences in member size. Irregularities concentrate damage in one place; the usual problems and their remedies are:

Irregularity	Why it is dangerous	Remedy
Soft / open ground storey (parking, shops)	Ground storey much weaker/more flexible than those above: all drift concentrates there and columns crush (pancake collapse).	Provide shear walls or infill in the ground storey; design columns for 2.5x the normal storey forces; avoid sudden stiffness change.
Short (captive) column	Partial-height infill or mezzanine makes a column short and very stiff: it attracts large shear and fails in brittle shear.	Avoid partial infill; isolate infill from columns; close-spaced ties over the full height.
Plan irregularity (L, T, U shapes)	Centre of mass and centre of rigidity do not coincide, causing torsion; re-entrant corners concentrate stress.	Use compact symmetric plans; separate wings with seismic joints; balanced placement of walls/columns.
Vertical discontinuity	Columns or walls stopped at upper floors (floating columns) interrupt the load path.	Keep all lateral elements continuous from foundation to roof.
Pounding between buildings	Adjacent buildings of different heights hit each other and cause local failure.	Provide seismic separation gap (about 0.4-0.5% of height).
Heavy cantilevers / overhangs	Vertical acceleration and large lever arms overstress them; heavy top masses increase forces.	Limit projections (< 1.2-1.5 m), keep parapets and water tanks light and well anchored.

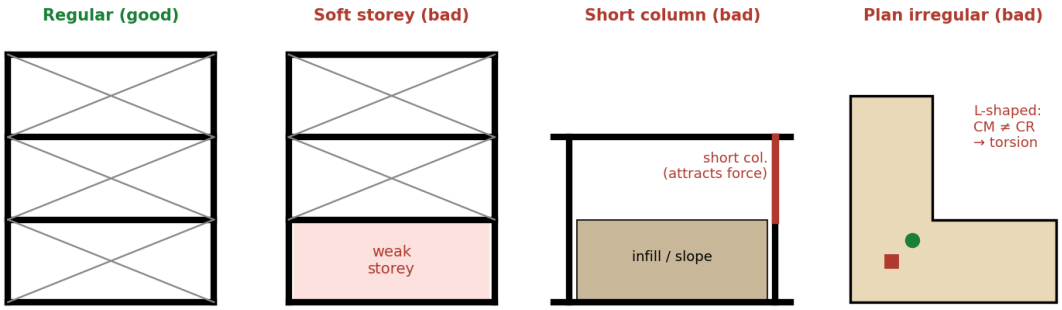


Fig. 3 Good and bad structural configurations (schematic)

5. Lateral load-resisting systems

System	How it resists earthquake forces	Typical use / remarks
Moment-resisting frame (MRF)	Rigid beam-column joints resist sway by bending; flexible, good ductility if detailed per IS 13920.	Low- to mid-rise residential and office buildings; large drift for tall frames.
Shear wall / dual system	RC walls act as vertical cantilevers with very high stiffness and strength; frames carry gravity and share lateral load.	Mid- to high-rise; most economical for 6-20 storeys; reduces drift and non-structural damage.
Braced frame (steel)	Diagonal members carry lateral load as axial tension/compression; eccentric bracing adds ductility.	Steel structures, retrofitting of existing frames.
Base isolation	Flexible bearings (lead-rubber or friction pendulum) lengthen the period and let the ground move under the building.	Hospitals, data centres, heritage; significantly reduces forces and damage but has higher initial cost.
Supplemental dampers	Viscous/metallic dampers dissipate seismic energy and reduce drift.	Tall or important structures, retrofit.

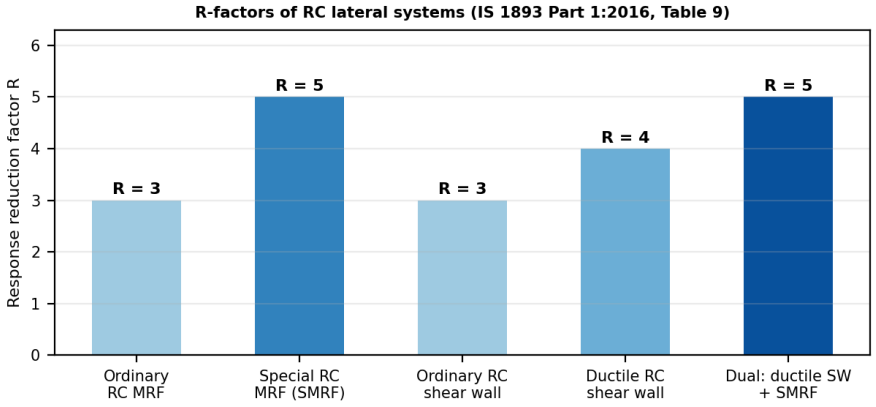


Fig. 4 Response reduction factors credited to RC systems in IS 1893 (R = 5 requires special ductile detailing)

6. Ductile detailing of RC members (IS 13920:2016)

Ductility is obtained from **confinement** of concrete by closely spaced hoops and from careful reinforcement arrangement. The most important rules are:

- **Beams:** width ≥ 200 mm and width/depth ≥ 0.3; tension steel not less than $0.24\sqrt{f_{ck}}/f_y$ (0.24% for M25/Fe500) and not more than 2.5%; at least two bars continuous at top and bottom; stirrups with **135° hooks**; in the plastic-hinge region (2d from the column face) stirrup spacing ≤ smaller of d/4, 8 x bar diameter and 100 mm.
- **Columns:** least dimension ≥ 300 mm; longitudinal steel 0.8%-4%; confining hoops at spacing ≤ 1/4 of the least dimension, between 75 and 100 mm, at both ends of the column and through the beam-column joint; lap splices only in the central half of the column height.
- **Joints:** hoops are continued through the joint core; beam bars are anchored with a 90° bend inside the column core; strong-column/weak-beam check $\Sigma M_c \geq 1.4\ \Sigma M_b$.
- **Quality:** cover and lap lengths as per code, concrete not below M20-M25, no honey-combing in the joints - poor workmanship can defeat good design.

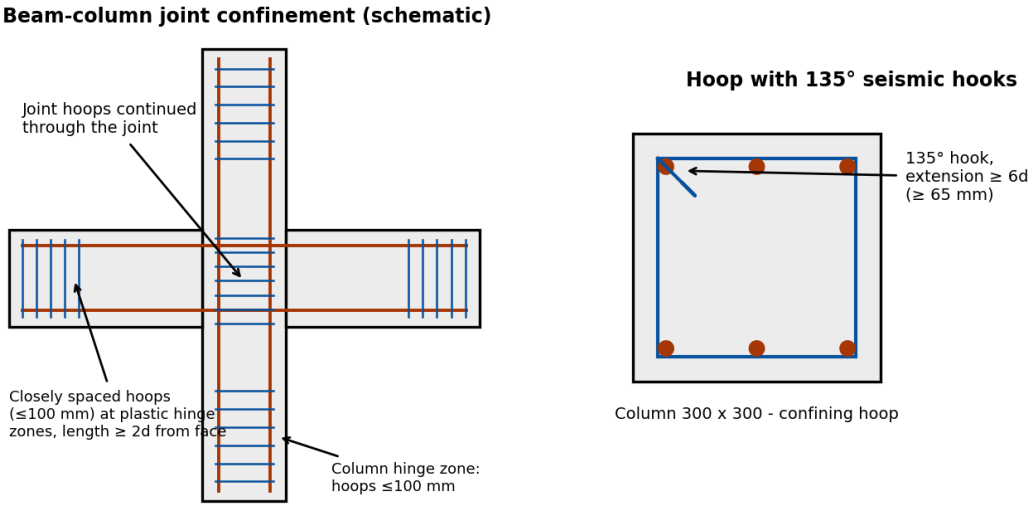


Fig. 5 Confinement at the beam-column joint and a hoop with 135° hooks

7. Case study: applying the concepts to the G+1 house (Tasks 1 and 2)

The house is a regular rectangular frame (plan aspect ratio 1.17, no re-entrant corners, walls on all sides of the ground storey, small 1.2 m balcony cantilever), so it already avoids the soft-storey, plan-irregularity and heavy-overhang problems. Seismic parameters: Zone IV (Z = 0.24), I = 1.0, R = 5 (SMRF), medium soil, T_a = 0.29 s, $S_a/g = 2.5 \rightarrow A_h = 0.060$. Seismic weight W = 2755 kN, so $\mathbf{V_B = 165\ kN}$ (about 6% of the weight), distributed as **37 kN** at the first floor and **128 kN** at the roof (Fig. 6). This is 2.3 times the wind force, so earthquake governs.

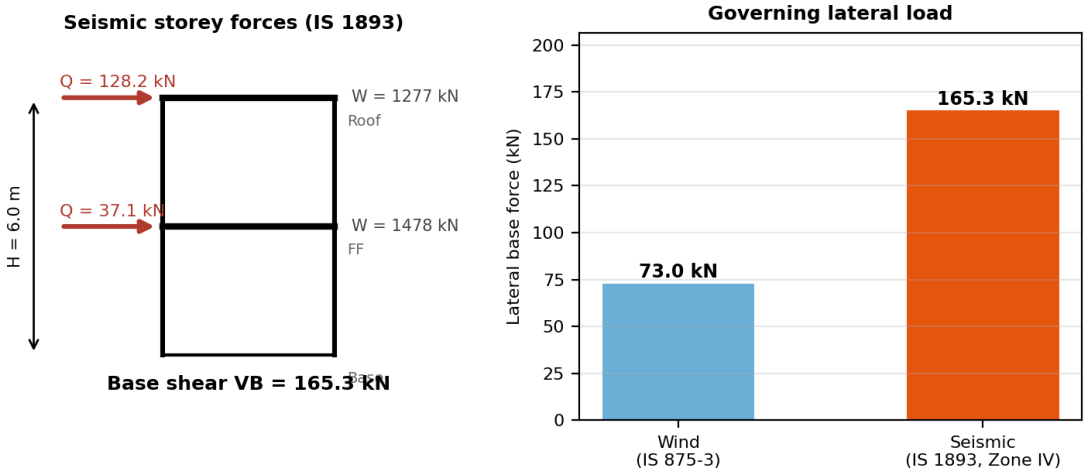


Fig. 6 Seismic storey forces on the case-study building and comparison with wind

Check	Gravity design (Task 2)	Seismic requirement	Recommendation
Beam 230 x 500	b/D = 0.46; 4-16φ bottom; 8φ @ 300	b ≥ 200, b/D ≥ 0.3 ✓; hinge-zone stirrups ≤ 100 mm	Use 8φ @ 100 within 2d (1.0 m) of each support with 135° hooks
Column 300 x 300	4-16φ; 8φ ties @ 250	Min. side 300 mm ✓; hoops 75-100 mm at ends and in joints	Use 8φ hoops @ 100 at both ends (≥ 500 mm) and in the joint; 135° hooks
Strong column / weak beam	Not checked in gravity design	ΣMc ≥ 1.4 ΣMb at every joint	With 230 x 500 beams, 300 x 300 columns are likely to fall short: verify in the frame model and enlarge columns (e.g. 300 x 450 or 400 x 400) if required
Brick infill walls	Treated as dead load only	Infill changes stiffness and can create short columns	Tie walls to frame with dowels; avoid partial-height infill beside columns
Staircase / mumty, tank	Included as mass	Heavy top masses increase roof force	Anchor tank on beams directly above columns; keep mumty walls light
Foundation	Not in scope	Uniform settlement; continuous load path	Isolated footings tied by plinth beams in both directions

8. Further concepts: new technologies and low-cost measures

- **Base isolation and dampers** decouple or dissipate energy; they are increasingly used for hospitals and schools in high-hazard regions.
- **Confined masonry and bands:** for low-cost housing, brick walls confined by RC tie-columns and horizontal bands (plinth, lintel and roof band) perform far better than unreinforced masonry.
- **Site and foundation:** avoid soft or liquefiable soil and steep slopes; do a soil investigation; tie all footings together.
- **Retrofitting** existing buildings using RC/steel jacketing, added shear walls or FRP wrapping is often more economical than reconstruction.
- **Non-structural safety:** anchor parapets, water tanks, ceilings and heavy furniture - falling objects cause many injuries.

9. Conclusions and recommendations

- Earthquake safety begins with a **simple, symmetric and continuous structural configuration**; the G+1 house in this project meets this.
- For the case-study building, seismic base shear (165 kN) is 2.3 times the wind force; earthquake, not wind, governs the lateral design.
- The assumed R = 5 is only valid with **special ductile detailing**: 135° hooks, 100 mm confining hoops in beam/column ends and joints.
- The **strong-column/weak-beam** condition should be verified, and column size increased if needed, before finalising the design.
- Good materials, supervision and workmanship are as important as calculations - a well-detailed, well-built simple building performs better than a complex one.

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